

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 HT	(a)	(i)		(Strong) attraction between positive ions (1) and (a sea of) {delocalised/free} electrons (1)	2			2		
		(ii)		Malleable(1) (Regular array of) atoms in layers that can easily slide over one another (1)	1	1		2		
		(iii)		{Free/Delocalised} electrons (1) are able to move (1)	2			2		
	(b)	(i)		Regular {pattern of atoms/layers} is 'broken' (1) so more difficult to make them slide (1)		2		2		
		(ii)	I	any 2 × (1) from: Lower density / lighter (1) and stronger (1) more flexible (1) Accept converse argument for steel			2	2		
			II	% of vanadium = 4% (1) 4% of 2 500 = 100 kg (1) volume = $\frac{100}{5700}$ (ecf)(1) substitution and manipulation = 0.0175/0.018 [m ³] (1) Do not accept 0.02 unless working shown Accept $\frac{2\,500}{5\,700}$ (1) =0.439 [m ³] (1)		4		4	4	
				Question 3 total 14	5	7	2	14	4	